

ELEC4631 Tutorial 11

1. Consider the controlled ODE given by

$$\begin{aligned}\dot{x}_1(t) &= -x_1(t)^3 + u(t), \\ \dot{x}_2(t) &= x_1(t).\end{aligned}$$

Using the candidate Lyapunov function $V(x_1, x_2) = x_1^2 + x_2^2$, find a scalar state-feedback control law $u(t) = \eta(x_1(t), x_2(t))$ such that the origin becomes *globally asymptotically stable*.

2. Consider the controlled ODE given by

$$\begin{aligned}\dot{x}_1(t) &= x_2(t), \\ \dot{x}_2(t) &= -x_2(t)|x_2(t)| + x_1(t)^3 + u(t).\end{aligned}$$

We want to drive the state to the origin and for it to come to rest there as $t \rightarrow \infty$. Using a Lyapunov function of the form

$$V(x_1, x_2) = \frac{1}{2}(x_2^2 + \lambda x_1^2),$$

for some constant $\lambda > 0$, propose a state-feedback control law u that will achieve this objective and show that the state does converge to the origin.

3. An example of a fully actuated mechanical system is an inverted pendulum with a torque applied at the base of the pendulum, see Fig. ?? . The equation of motion for the pendulum is given by the ODE:

$$ml^2\ddot{\theta} = mgl \sin \theta - b\dot{\theta} + T,$$

where $m > 0$ is the mass of the pendulum, $g > 0$ is the gravity's acceleration and $b > 0$ is a damping coefficient. So, the equation is of the form presented in Lecture 4 with $q := \theta$, $F := T$, $M(q) := ml^2$, $B(q) := b$, and $G(q) := -mgl \sin \theta$.

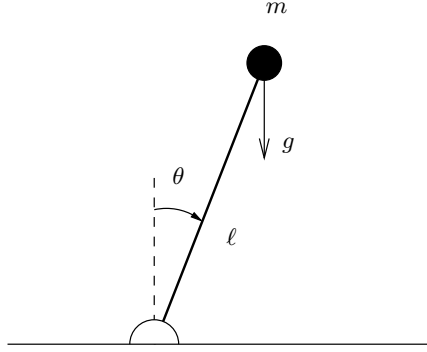


Figure 1: An inverted pendulum with a torque at the base.

Defining the state variables $x_1 = q$ and $x_2 = \dot{q}$, the equation of motion can be written as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ \frac{g}{l} \sin x_1 - \frac{b}{ml^2} x_2 + \frac{T}{ml^2} \end{bmatrix}.$$

Let $u = T$. Using the Lyapunov function $V(x_1, x_2) = \frac{1}{2}(x_1^2 + ml^2 x_2^2)$, design a state-feedback control law u such that $x_1(t), x_2(t) \rightarrow 0$ as $t \rightarrow \infty$ and prove that the control objective is attained by this law.

4. Consider a system on the state-space $S = \mathbb{R}^2$ described by the ODE

$$\begin{aligned} \dot{x} &= \omega(x^2 + y^2)y - (x + 4y)(x^2 + y^2 - 1) \\ \dot{y} &= -\omega(x^2 + y^2)x + (4x - 5y)(x^2 + y^2 - 1), \end{aligned}$$

with $\omega > 0$. Let $C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. Consider $\Omega_{r,c} = \{(x, y)^T \in \mathbb{R}^2 \mid \|(x, y)^T - z\| \leq r \text{ for some } z \in C, V(x) \leq c\}$ with $r < 1$ and $c > 0$ such that it is positively invariant. Using the candidate Lyapunov function

$$V(x, y) = \frac{1}{4}(x^2 + y^2 - 1)^2,$$

show that all trajectories starting in $\Omega_{r,c}$ asymptotically converges to the set C .